

[SCHOOL NAME — PLACEHOLDER LETTERHEAD]

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Class 3 Maths — CA-1 (Hard)

Class : 3

Subject : Maths (Math-Mela)

Max Marks : 30

Duration : 1 Hour

Syllabus: Chapters 1–3 — What's in a Name?, Toy Joy, Double Century

General Instructions: All questions are compulsory. For Section A, write the correct option letter.

SECTION A — Multiple Choice Questions (1 Mark each)

- A basket has 187 mangoes. If they are packed into boxes of 10, how many full boxes can be made and how many mangoes are left over? (1 Mark)
A. 17 boxes, 7 left over
B. 18 boxes, 7 left over
C. 18 boxes, 0 left over
D. 19 boxes, 7 left over
- Which shape, when cut straight through the middle parallel to its base, always shows a circle as the cut surface? (1 Mark)
A. Cube
B. Cuboid
C. Cylinder
D. None of these
- A cuboid has length, breadth, and height all different from each other. How many pairs of identical (equal) faces does it have? (1 Mark)
A. 1
B. 2
C. 3
D. 6
- If the number 1_5 (tens digit missing) is greater than 143 but less than 168, which digit could go in the blank? (1 Mark)
A. 3
B. 5
C. 2
D. 9
- What is the greatest 3-digit number you can make that is less than 200 using the digits 1, 9, and 8 (using each digit exactly once)? (1 Mark)
A. 198
B. 189
C. 918
D. 981
- A toy is made by joining a cone on top of a cylinder. How many total flat faces does the combined toy have (counting only the outer visible flat faces)? (1 Mark)
A. 1
B. 2
C. 3
D. 4
- Which of these statements is TRUE about all cubes? (1 Mark)
A. A cube is never also a cuboid
B. Every cube is a special type of cuboid
C. A cube has curved surfaces
D. A cube has only 4 faces

SECTION B — Short Answer Questions (2 Marks each)

- A pond has 176 fish. If 38 fish are caught and 22 new fish are added, how many fish are in the pond now? (0 Marks)
- Explain why a sphere cannot be stacked like a cube, using the idea of flat and curved surfaces. (0 Marks)
- Write a 3-digit number where the hundreds digit is double the tens digit, and the ones digit is 5. Explain your reasoning. (0 Marks)
- Two toy blocks, a cube and a cuboid, look similar from one side. Explain one way you could tell them apart just by looking at all their faces. (0 Marks)
- A number lies between 150 and 200. Its tens digit is 7 and its ones digit is the same as its tens digit. What is the number? (0 Marks)
- A rectangular toy box (cuboid) is placed so its largest face touches the ground. Explain why this makes the box more stable than standing it on its smallest face. (0 Marks)

SECTION C — Short Answer Questions (Explain) (3 Marks each)

- A trader has 200 oranges. Every day, the trader sells some oranges and this pattern continues: Day 1 sells 34, Day 2 sells 12 more than Day 1, and Day 3 sells 8 fewer than Day 2. Find the total oranges sold over the three days and the oranges left after Day 3. (0 Marks)
- Explain, using properties of faces, edges and corners, how you would prove to a younger student that a cube and a cuboid are related but not exactly the same shape. (0 Marks)

SECTION D — Long Answer Questions (5 Marks each)

- A large box contains 200 building blocks made up of cubes, cuboids, cylinders, cones, and spheres. If there are 40 cubes, 35 cuboids, 45 cylinders, and 30 cones, find how many spheres are in the box, and then explain one property that makes spheres different from all the other four shapes in this box. (0 Marks)

